

COURSE DESCRIPTION			
Program	Chemical Engineering		
Course Title	Fundamentals of Engineering Chemistry		
Course Code	2071	Semester	II
Type of Course	Practicum	Contact Hours	90
Teaching Scheme (L: T:P:R)	3:0:3:0	Credits	4.5
CIE Marks	40	ESE Marks	100

Introduction

Fundamentals of Engineering Chemistry course introduces fundamental concepts of electrochemistry, surface chemistry, colloids, and heterogeneous systems. It also covers the basic concepts of organic chemistry, some industrially important organic compounds, their reactions and the practical sessions enhance the understanding through experiments.

Course Objectives

1	To impart basic knowledge of electrochemistry and corrosion and enable students to solve problems in engineering systems.
2	To provide a fundamental idea on surface chemistry, colloids and heterogeneous systems.
3	To enable students to know the basic concepts of organic chemistry.
4	To equip students with some important organic compounds and reactions.

Course Pre-requisite

Topic	Course code	Course Title	Semester
Basic concepts of Chemistry	2003B	Chemistry for Technical Practices	I

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Apply the basic knowledge of electrochemistry and corrosion and enable students to utilize it in engineering systems to solve industrial challenges.	Apply
CO2	Apply the basic concepts and principles of adsorption mechanisms, adsorption isotherms, various colloids and heterogeneous systems.	Apply
CO3	Apply the fundamental concepts of organic chemistry to solve elementary problems in engineering chemistry.	Apply
CO4	Explore various industrially important organic compounds and their reactions needed to develop the ability to solve engineering problems.	Apply

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	-	-	-	-	-	3	3	3	3
CO2	3	3	-	-	-	-	-	3	3	3	3
CO3	3	3	-	-	-	-	-	3	3	3	3
CO4	3	3	-	-	-	-	-	3	3	3	3
Total	12	12	-	-	-	-	-	12	12	12	12
Correlation Level	3	3	-	-	-	-	-	3	3	3	3

Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme						
							Theory			Practical			Total
L	T	P	S				CIE	ESE	Total	CIE	ESE	Total	
3	0	3	0	6	90	4.5	20	60	80	20	40	60	140

L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination

Syllabus - Major Topics

Module	Title	Major Topics	Instructional Hours	
			L	P
Module I	Electrochemistry, Corrosion and Emerging Battery Technologies	Electrochemistry: Oxidation, reduction and redox reactions - conductors and insulators - metallic conductors and electrolytic conductors - strong electrolytes, weak electrolytes and non-electrolytes - electrolysis, electrolytic cell - Faraday's laws of electrolysis - simple problems based on Faraday's laws - Industrial applications of electrolysis: electroplating and electrolytic refining - Electrochemical cell (Daniel cell) - classification: Primary cell, secondary cell and fuel cell. Corrosion of metals - Factors affecting rate of corrosion - Corrosion prevention methods: a) Barrier protection-metallic (anodic, cathodic) coatings, non-metallic (anodising and anti-rust solution) b) Cathodic protection- sacrificial anode method. Emerging battery technologies (Basic idea only) with the following examples: Li-ion, Na-ion, solid state battery.	11	0
Module I	Experiments based on electrochemical cell conductivity and electrolysis	Construction of electrochemical cell, conductivity measurement and electroplating.	0	9
Module II	Surface chemistry, Colloids and Heterogeneous systems	Surface Chemistry - Adsorption- types of adsorption of gases by solids, the factors influencing adsorption- Freundlich and Langmuir adsorption isotherm (derivation not required). Colloids- True solution, colloidal solution and suspension - Classify colloids to lyophilic, lyophobic, macromolecular, multimolecular and associated colloids- properties of colloids: Brownian movement, Tyndall effect and electrophoresis - applications of colloids: delta formation, medicines and emulsification- Purification of colloids by dialysis, electrodialysis, ultrafiltration and ultracentrifugation. Heterogeneous Systems-phase, components and degrees of freedom of a system- Gibb's phase rule and its equation- application of Gibb's Phase Rule to one component system (water system with phase diagram).	11	0
Module II	Experiments based on adsorption and colloids	Adsorption experiments and experiments on colloids.	0	9
Module III	Introduction to Chemical bonding, Unique properties of carbon, and Methods of purification of organic compounds	Chemical bond: ionic, covalent and co-ordinate bond with one example each. Unique nature of carbon with reference to tetra valency, catenation and allotropes. Hybridization - sp, sp ² , and sp ³ hybridization taking examples of methane, ethene and ethyne. IUPAC Nomenclature of organic compounds with single functional group only. (Straight chain, branched and cyclic hydrocarbons, compounds having -OH, -COOH, -CHO, -C=O, -CONH ₂ -X, -OR, -COOR, -NO ₂ and -NH ₂). Purification of organic compounds-Sublimation, crystallisation, distillation, fractional distillation, differential extraction and chromatography (only basic idea of all methods).	11	0
Module III	Organic Chemistry Experiments	Identification of organic compounds containing various functional groups (-OH, -COOH, -CHO, -C=O, -CONH ₂ , -X, -OR, -COOR, -NO ₂ and -NH ₂), preparation of hydrocarbon derivative(picric acid).	0	9

Module IV	Preparation and chemical reactions of some important organic compounds	Preparation of hydrocarbons—methane by decarboxylation of carboxylic acid using sodalime distillation—ethane by Wurtz reaction of chloromethane—ethene by dehydrohalogenation of chloroethane using alcoholic KOH—ethyne by action of water on calcium carbide. Chemical reactions of hydrocarbons - chlorination of methane, ethane and ethyne- Addition reaction of Br₂ in CCl₄ to alkenes (test of unsaturation) - Addition of HBr to symmetrical and unsymmetrical alkenes - Markovnikov's and AntiMarkovnikov's addition Acidity of ethyne - reaction with sodium and sodamide (test of alkynes) : Uses of methane, ethane, ethene and ethyne. Aromatic Compounds : Preparation of benzene by cyclic polymerization of ethynes- Friedel - Craft's alkylation reaction(Friedel - Craft's methylation of benzene)and Friedel -Craft's acylation(Friedel-Craft's benzylation of benzene), Uses of benzene. Preparation of alcohols: Preparation of ethanol—fermentation reaction and action of Grignard reagent on methanal - reaction of ethanol with ethanoic acid (esterification)—reaction of ethanol with chlorinating agents like Luca's reagent, PCl₃, PCl₅ and SOCl₂- oxidation reaction of ethanol using PCC and acidified K₂Cr₂O₇ or KMnO₄ - two uses of ethanol. Preparation of phenol - Dow's process (from chlorobenzene) - reactions of phenol: bromination (Br₂ in CS₂ and Br₂ in water) and nitration (dil. HNO₃ and Conc.HNO₃)- (no mechanism is required)- Kolbe reaction of phenol. Method of preparation of ethanoic acid from CO₂ using Grignard reagent - chemical reactions: reduction reaction and HVZ reaction of ethanoic acid.	12	0
Module IV	Organic Chemistry Experiments	Reaction of ethanol with ethanoic acid (esterification), Lucas' test on ethanol, Friedel Crafts benzylation of benzene, preparation of 5- nitro salicylic acid from salicylic acid..	0	12

Detailed Syllabus

Subtopic	Student Learning Outcome	Mode of delivery (L/T)	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
Electrochemistry	Define oxidation, reduction, redox reactions, conductors and insulators with suitable examples for each.	L	CO1	Remember	1
Electrochemistry	Differentiate between metallic conductors (electronic conductors) and electrolytic conductors (ionic conductors) with examples.	L	CO1	Understand	1
Electrochemistry	Define electrolytes (strong and weak electrolytes), non-electrolytes, electrolysis and electrolytic cell with suitable examples.	L	CO1	Remember	1
Electrochemistry	State Faraday's laws of electrolysis (First and second law).	L	CO1	Remember	1
Electrochemistry	Solve problems based on Faraday's laws(First and second law).	L	CO1	Apply	1
Electrochemistry	Illustrate industrial applications of electrolysis—electroplating (electroplating of Nickel on mild steel) and electrolytic refining (electrolytic refining of copper).	L	CO1	Understand	1
Electrochemistry	Illustrate electrochemical cell using Daniel cell as example and also explain salt bridge.	L	CO1	Understand	1
Electrochemistry	Classify electrochemical cell into primary cell, secondary cell and fuel cell with one example for each.	L	CO1	Understand	1
Corrosion	Define corrosion of metals and list factors affecting rate of corrosion of metals.	L	CO1	Remember	0.5
Corrosion	Explain corrosion prevention methods: a) Barrier protection-metallic (anodic, cathodic) coatings, non-metallic (anodising and anti rust solution) b) Cathodic protection- sacrificial anode method.	L	CO1	Understand	2
Emerging battery technologies	Discuss about emerging battery technologies (Basic idea only) with the following examples: Li-ion, Na-ion, solid state battery.	L	CO1	Understand	0.5
Experiments based on electrochemical cell conductivity and electrolysis	Construct salt bridge/Construct Daniel cell.	P	CO1	Apply	3

Experiments based on electrochemical cell conductivity and electrolysis	Demonstrate the conductivity of a given water sample using a conductivity meter/Electroplating—Coating a steel spoon with Nickel (mixture of NiCl_2 and NiSO_4 as electrolyte.	P	CO1	Apply	3
Experiments based on electrochemical cell conductivity and electrolysis	Demonstrate electrolysis of water.	P	CO1	Apply	3
Module II					
Surface Chemistry	Differentiate types of adsorption of gases by solids (physisorption and chemisorption) with examples and list the factors influencing adsorption.	L	CO2	Understand	1.5
Surface Chemistry	Discuss Freundlich and Langmuir adsorption isotherm (derivation not required).	L	CO2	Understand	1.5
Colloids	Define true solution, colloidal solution and suspension.	L	CO2	Remember	0.5
Colloids	Classify colloids to lyophilic, lyophobic, macromolecular, multimolecular and associated colloids with examples.	L	CO2	Understand	1.5
Colloids	Explain the properties of colloids: Brownian movement, Tyndall effect and electrophoresis.	L	CO2	Understand	1.5
Colloids	List applications of colloids: Delta, formation, medicines, emulsification and discuss the purification of colloids by dialysis electrodialysis, ultrafiltration and ultracentrifugation.	L	CO2	Understand	2
Heterogeneous systems	Define phase, components, degrees of freedom - the fundamental variables to describe the state of a system and state Gibb's phase rule and give its equation.	L	CO2	Remember	1
Heterogeneous systems	Explain the application of Gibb's Phase rule to one component system (water system with phase diagram).	L	CO2	Understand	1.5
Experiments based on adsorption and colloids	Estimate amount of oxalic acid adsorbed by charcoal.	P	CO2	Apply	3
Experiments based on adsorption and colloids	Prepare two lyophilic colloids.	P	CO2	Apply	3
Experiments based on adsorption and colloids	Prepare two lyophobic colloids.	P	CO2	Apply	3
Module III					
Introduction to Chemical bonding, Unique properties of carbon, and Methods of purification of organic compounds.	Discuss about chemical bond (ionic,covalent and co-ordinate bond) with one example each(Examples: ionic- NaCl , covalent- HF , co-ordinate- NH_4^+ ion).	L	CO3	Understand	2
Introduction to Chemical bonding, Unique properties of carbon, and Methods of purification of organic compounds.	Explain the unique property of carbon with reference to tetra valency and catenation.	L	CO3	Understand	0.5
Hybridization	Illustrate hybridization (sp , sp^2 , and sp^3) taking examples of methane(sp^3) ethene (sp^2)and ethyne(sp).	L	CO3	Understand	2
IUPAC Nomenclature	Discuss IUPAC Nomenclature of organic compounds with single functional group only. (Straight chain, branched and cyclic hydrocarbons (aliphatic and aromatic), compounds having -OH, -COOH, -CHO, -C=O, -CONH ₂ , -X, -OR, -COOR,-NO ₂ and -NH ₂).	L	CO3	Understand	3
Purification methods of organic compounds	Explain various methods of purification of organic compounds- Sublimation, crystallisation, distillation, fractional distillatio, differential extraction and chromatography (TLC and column)-(only basic idea of all methods).	L	CO3	Understand	3.5
Organic Chemistry Experiments	Identify the organic compound with various functional groups. (-OH, -COOH, -CHO, -C=O, _CONH ₂)	P	CO3	Apply	3
Organic Chemistry Experiments	Identify the organic compound with various functional groups (-X, -OR, -COOR,-NO ₂ and -NH ₂ .)	P	CO3	Apply	3

Organic Chemistry Experiments	Prepare hydrocarbon derivative(picrate).	P	CO3	Apply	3
Module IV					
Important organic compounds: preparation and their chemical reactions	Explain the preparation method of methane by the decarboxylation of carboxylic acid using sodalime distillation and list any two uses of methane.	L	CO4	Understand	0.5
Important organic compounds: preparation and their chemical reactions	Explain the preparation method of ethane by Wurtz reaction of chloromethane and list any two uses of ethane.	L	CO4	Understand	0.5
Important organic compounds: preparation and their chemical reactions	Discuss the preparation of ethene by dehydrohalogenation of chloroethane using alcoholic KOH and list any two uses of ethene.	L	CO4	Understand	0.5
Important organic compounds: preparation and their chemical reactions	Explain the preparation of ethyne by action of water on calcium carbide and list any two uses of ethyne.	L	CO4	Understand	0.5
Important organic compounds: preparation and their chemical reactions	Explain chlorination of methane, ethane and ethyne.	L	CO4	Understand	1
Important organic compounds: preparation and their chemical reactions	Explain the addition reaction of Br ₂ in CCl ₄ to alkenes (test of unsaturation).	L	CO4	Understand	0.5
Important organic compounds: preparation and their chemical reactions	Discuss the addition of HBr to symmetrical (ethene and butene) and unsymmetrical (propene) alkenes –state Markovnikov's and Anti-Markovnikov's rule.	L	CO4	Understand	1
Important organic compounds: preparation and their chemical reactions	Explain the reason for acidity of ethyne, reactions showing acidity of ethyne: reaction with sodium and sodamide (test of alkynes).	L	CO4	Understand	0.5
Important organic compounds: preparation and their chemical reactions	Discuss the preparation of benzene by cyclic polymerization of ethyne.	L	CO4	Understand	0.5
Important organic compounds: preparation and their chemical reactions	Discuss Friedel-Crafts alkylation reaction (Friedel – Craft's methylation of benzene) and Friedel-Crafts acylation reaction (Friedel – Craft's benzylation of benzene) and list any two uses of benzene.	L	CO4	Understand	1
Important organic compounds: preparation and their chemical reactions	Explain the preparation of ethanol by i) fermentation reaction of molasses ii) the action of Grignard reagent on methanal.	L	CO4	Understand	1
Important organic compounds: preparation and their chemical reactions	Explain the esterification reaction of ethanol with ethanoic acid.	L	CO4	Understand	0.5
Important organic compounds: preparation and their chemical reactions	Discuss the reaction of ethanol with chlorinating agents like Luca's reagent , PCl ₃ , PCl ₅ , and SOCl ₂ .	L	CO4	Understand	0.5
Important organic compounds: preparation and their chemical reactions	Explain the oxidation reaction of ethanol using acidified K ₂ Cr ₂ O ₇ or KMnO ₄ and PCC and list any two uses of ethanol.	L	CO4	Understand	0.5
Important organic compounds: preparation and their chemical reactions	Discuss the preparation of phenol by Dow's process.	L	CO4	Understand	0.5
Important organic compounds: preparation and their chemical reactions	Discuss the bromination reaction of phenol (Br ₂ in CS ₂ and Br ₂ in water) -(no mechanism is required).	L	CO4	Understand	0.5
Important organic compounds: preparation and their chemical reactions	Discuss nitration of phenol with dil. HNO ₃ and Conc. HNO ₃ —(no mechanism is required).	L	CO4	Understand	0.5
Important organic compounds: preparation and their chemical reactions	Explain Kolbe reaction for the synthesis of salicylic acid.	L	CO4	Understand	0.5
Important organic compounds: preparation and their chemical reactions	Explain the method of preparation of ethanoic acid from CO ₂ using Grignard reagent.	L	CO4	Understand	0.5
Important organic compounds: preparation and their chemical reactions	Explain the reduction reaction of ethanoic acid with LiAlH ₄ and HVZ reaction of ethanoic acid.	L	CO4	Understand	0.5

Organic Chemistry Experiments	Lucas' test on ethanol.	P	CO4	Apply	3
Organic Chemistry Experiments	Prepare an ester of ethanol with ethanoic acid (esterification)	P	CO4	Apply	3
Organic Chemistry Experiments	Friedel Crafts benzylation of benzene.	P	CO4	Apply	3
Organic Chemistry Experiments	Prepare 5-nitrosalicylic acid from salicylic acid.	P	CO4	Apply	3

Suggested Student Activities for Self-Learning

Week	Self-Learning description	Hours
Module I		
1	- Write a report on hydrogen economy (use of hydrogen fuel in automobiles). - Completion of record work and result analysis.	6
2	- Prepare an assignment on various oxidation-reduction reactions and explain the working of Daniel cell with diagram. - Completion of record work and result analysis.	6
3	- Write a report on recycling of batteries. - Completion of record work and result analysis.	6
4	- Prepare a report on types of batteries used in electric vehicles and also explain the advantages and disadvantages of electric vehicle batteries - Completion of record work and result analysis.	6
Module II		
5	- Prepare a content (video content, chart) on application of adsorption such as water purification. - Completion of record work and result analysis.	6
6	- Identify the different colloidal systems in day today life. - Prepare minimum three different types of lyophilic sols. - Completion of record work and result analysis.	6
7	- Demonstrate tyndall effect/brownian movement by simple experiment. - Study minimum three effects due to Tyndall effect. - Completion of record work and result analysis.	6
8	- Draw a one component phase diagram and explain. - Preparation of series Test, submission of assignment based on CO1 & CO2. - Completion of record work and result analysis.	6
Module III		
9	- Write an assignment on various purification methods of organic compounds.. - Completion of record work and result analysis.	6
10	- Prepare an assignment on various types of bonding with two examples each. - Completion of record work and result analysis.	6
11	- Prepare an assignment on unique properties of carbon atom - Completion of record work and result analysis.	6
12	- Explain catenation of carbon element which gives rise to linear, branched and cyclic organic compounds with examples. - Completion of record work and result analysis.	6
Module IV		
13	- Find any three industries which manufacture or utilize alcohols, phenols, aldehydes and ketones. - Completion of record work and result analysis.	6
14	- Identify ten organic chemicals containing products used in household activities. - Completion of record work and result analysis.	6
15	- Take three personal care and hygiene products (sanitizers, perfumes, and disinfectants) and find the constituents present in that products. - Completion of record work and result analysis.	6
Total Self-Learning hours		90

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
1	Principles of Physical Chemistry	Puri, Sharma & Pathania	Vishal Publishing.Co.

2	Textbook of Physical Chemistry	P.L.Soni, O.P.Dharmarha , U.N.Dash	Sultan Chand & Sons, New Delhi
3	Organic Chemistry	Bahl & Arun Bahl	S. Chand & Company Ltd.
4	Principles and Applications of Electrochemistry	D.R. Crow	CRC Press/Chapman & Hall

REFERENCE			
Sl. No.	Title	Author	Publication
1	Physical Chemistry	Atkins	
2	Organic Chemistry	Morrison &Bohd	

WEB RESOURCES		
Sl. No.	URL	Modules Covered
1	https://www.shaalaa.com/concept-notes/organic-chemistry_6527	All
2	https://nptel.ac.in/courses/104106119	All

Major Equipment/Instruments/Components required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
1	Conductivity meter	Digital LCD (e.g., digit) displaying conductivity,;benchtop models (AC power, RS232 output)	1
2	Laboratory mechanical shaker	Automatic	1
3	Bunsen burner		2
4	Digital weighing balance		1

Major Software required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
	NIL		

Assessment Methodology - Practicum

Mode	Attendance	CA1	CE	CA2	CA3	CA4	CA5	ESE	
		Average of Self learning activities from each module	Continuous evaluation	Practical test (first half of experiments)	Practical test (second half of experiments)	Series Exam (2 modules)	Series Exam (2 modules)	Written Exam (theory)	Lab Examination (internal)
Duration				3 hours	3 hours	2 hours	2 hours	2.5 hours	3 hours
Exam mark		10	50	40	40	50	50	60	40
Converted to		5	5	10	10	10	10		
Marks	5	5	5	10		10		60	40
Total	40							100	
Tentative schedule	End of semester	By 4th week	By 7th week	By 11th week	By 14th week	8th week	15th week	End of semester	

Scheme of Evaluation - Practical (Continuous Internal Evaluation)

Sl. No.	Criteria	Description	Marks
1	Preparation & Punctuality	Attendance, timely submission of record, pre-lab preparation, understanding of experiment objectives and theory.	10
2	Experimental Setup & Procedure	Ability to identify components/equipment, correct circuit connections/setup, systematic and safe execution of the procedure.	10
3	Observation & Data Recording	Accuracy and neatness of readings, correct use of units, tabulation, and systematic recording of results.	5

4	Analysis & Result Interpretation	Proper calculations, graphical representation, result verification, discussion, and logical conclusion.	10
5	Viva Voce / Concept Understanding	Clarity of theoretical concepts, explanation of procedure, ability to answer questions related to the experiment.	10
6	Workmanship, Discipline & Teamwork	Laboratory discipline, care of instruments, teamwork, attitude, and safety practices.	5
Total			50

Scheme of Evaluation - Practical Test

Category	Things to evaluate	Marks
Write-up / Procedure (before experiment)	Aim, circuit diagram/flowchart, stepwise procedure.	10
Experiment Setup & Execution	Correct circuit connections / coding / setup. Handling of instruments/software. Accuracy of execution.	10
Observation & Result / Output	Proper tabulation of readings. Calculations/graphs/program output. Correctness of result / expected outcome.	8
Viva Voce	Conceptual understanding of experiment. Related theoretical knowledge.	8
Record	Completion & neatness of practical record. Proper certification by faculty.	4
Total		40

Series Examination Pattern- Theory

Time	Module	Part A	Marks	Part B	Marks	Part C (With choice)	Marks	Total
2 hours	Any one module	2	1	3	3	2	7	25
	Any one module	2	1	3	3	2	7	25
	Number	4		6		4		50

End Semester Examination Pattern of Question Paper - Theory

Duration	Module	TYPE OF QUESTIONS						Total Marks
		Part A (2 Questions from each module)		Part B 2 out of 3 questions from each module)		Part C (1 set questions with choice from each module)		
		No. of Questions	Marks (1 mark each)	No. of Questions	Marks (3 marks each)	No. of Questions	Marks (7 marks each)	
2.5 hours	1	2	2	3	6	1 set	7	15
	2	2	2	3	6	1 set	7	15
	3	2	2	3	6	1 set	7	15
	4	2	2	3	6	1 set	7	15
		8	8	12	24	4 sets	28	60